R23.2b High-wind forecasting and active control for buildings

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 14:35 — 2026-10-14 16:49

Stage conclusion

Project report

A2 High-wind forecasting and active control for buildings

Objective

Across three control layers — forecast-driven (hours–days) control mode switching, second–minute feedforward, and real-time feedback — determine which layer AI weather forecasts can serve; control encompasses active mass dampers (AMDs) and active aerodynamic shaping, with wind as the unified input.

Current Status

Node opened on 10-14. 14:38 Step 1: Grok quick search dispatched and returned (14:39, report, abstract-level only); 14:5x Step 2 Deep Research (--gemini) dispatched (brief), borrowing the feedforward conclusions from

A3 Grok report cb8bff without re-searching (its Section 6 numbers were later judged unverified by A3 review and discarded, replaced by citations verified in A3). Original plan: (brief), read A3's Grok report after our report returns (borrowing the feedforward portion), then draft the second Deep Research (--gemini) brief, and send conclusions to

Deep Research returned at 15:0x; after Lead verified citations, sent to review **** (3 reviewers, dispatched at 15:1x, awaiting results). Review packet.

Review (E with reservations, A does not support, D with reservations) rendered verdict →. Original minimal experiment (AI forecast → TPU → threshold exceedance) cancelled. New minimal experiment dispatched to ideas **** (queued): in-house Silsoe LBM lead-time τ prediction + stroke-limited active mass damper preview feedforward, workstation CPU, 0 CU, task brief.

Three-Layer Table (Finalized after Review)

Time Scale	Control Action	Required Lead Time and Accuracy	Precedents (Citation & Page)	Can AI Forecasts Supply It?
Hours–Days	Slow actions only: mass-block recentering and mechanical unlocking/locking, hydraulic/backup power preparation and manual check, slow mechanical aerodynamic shaping or TLD water filling/draining, full-storm stroke budgeting	Literature does not specify execution duration or latest decision time; "requires hour-scale lead time" is an assumption	No written operating procedures for "switching based on weather forecasts" (none found across three independent searches; not claiming non-existence); commissioned devices trigger in real time via acceleration/stroke thresholds (PAFC: Zhou & Li 2022 Eng Struct 269:114840 abstract, "accurately triggered according to operation principles", damping ratio 0.6%→2–7%, no threshold values given)	Yes, but marginal value is small: slow actions are already covered by official typhoon/hurricane warnings; global AI models lack skill for hurricane intensity (DeMaria 2025 card, event_step0, line 14) → **deferred**

Time Scale	Control Action	Required Lead Time and Accuracy	Precedents (Citation & Page)	Can AI Forecasts Supply It?
Seconds–Minutes	Near-field preview feedforward / preview MPC; gain reduction, semi-active gear shifting (millisecond–second scale, real-time sensing suffices)	Preview seconds: no verified source (numbers in Section 6 of A3 Grok report discarded after A3 review); wind-turbine lidar feedforward is effective under stationary turbulence: Guo 2023 WES 8:149 §5.2 tower-base fore-aft bending moment DEL up to −16.7%, Bossanyi 2014, Schlipf 2018 (verified by A3)	No full-scale engineering validation; research/concept level: Abdelaziz & Hobeck 2022 SPIE 12046 (10.1117/12.2630805, surrounding-building pressure taps + GRU prediction window + aerodynamic corner flaps, 1:400 wind tunnel, RMS torsional acceleration up to −75%, title verified, content cited via reviewer D); US5592791A (1997 patent, corner outrigger boom upstream wind measurement feedforward, original text verified by Lead); Klein 1972, Soong & Skinner 1981, Abdel-Rohman 1983 synchronous wind-sensing aerodynamic appendages (cited via Korkmaz 2011 journal p. 2130); theoretical Marzbanrad 2004 Eng Struct pp.1463–1471	No — requires on-site near-field wind measurements; **narrowed focus of this node**; in-house LBM τ=0 correlation 0.67 at 7 m → 0.35 at 27 m (in-house data)
Real-time	Acceleration/displacement feedback	None	Standard practice	Not needed

Existing preview statistics results: near-field velocity provides second-scale preview of windward Cp (correlation half-decay lead time 2.6–6.5 s, in-house data). Control Round 2 ideas (dispatched 10-14 16:1x) first performs a control-model self-check: without saturation, perfect preview must reduce acceleration RMS by ≥30%, before measuring the ratio of predicted-preview gain to perfect-preview gain; task brief, deliverables.

One-sentence conclusion (finalized 10-14): Shelved. AI weather forecasts can only serve hour-scale slow actions with small marginal value; second-scale near-field preview offers large theoretical gains (perfect preview reduces acceleration by 68–80%), but prediction from sparse upstream probes captures only 0–17% of that gain, with the bottleneck in inflow prediction skill. For causes, missing conditions, and restart criteria, see
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(Earlier provisional note) Driving building control with AI weather forecasts has low value, so that layer is shelved; what is worth pursuing is second-scale near-field preview feedforward (research-level precedents exist, no full-scale engineering validation), first using in-house data to quantify usable lead time and gain for stroke-limited dampers before deciding whether to promote or merge into A3's downburst preview experimental framework.

Established Decisions

- 2026-10-14 Literature conclusion verdict and scope narrowing →

Experiments Conducted and Results

Date	What Was Done	One-Sentence Result
10-14	Grok quick search	Almost no precedents for building/bridge wind-induced vibration control driven by "upstream inflow preview" or "numerical weather prediction"; existing "prediction" studies mostly involve MPC state extrapolation / disturbance estimation; operational ATMDs start up and lock based on acceleration thresholds (fragmentary numbers only, unverified). Three issues: 2004 preview control DOI actually points to another paper (Xie 2004 interference effects); Taipei 101 TAO DOI not found; "~0.75–1.5 s effective preview" was back-calculated by Grok from our in-house 30 m figure and does not count as literature evidence
10-14	Minimal experiment Round 2 v3 (ideas, workstation CPU 20.5 s, 0 CU)	With corrected feedforward, perfect 8 s preview reduces acceleration RMS by 68% (near spectral peak) / 80% (off-resonance); velocity-probe predicted preview captures only 0–17% of the perfect gain, with only 4 of 360 combinations (near spectral peak with tight stroke limit, 10.3–12.6% reduction) passing the gate, showing weak discriminative power; best off-resonance gain is 0.8% → bottleneck lies in inflow prediction skill

Date	What Was Done	One-Sentence Result
10-14	Minimal experiment v2 (ideas, workstation CPU 14.5 s, 0 CU)	Preview statistics: lead time at which windward 11-point Cp correlation drops by half is 2.6 s (7 m upstream) → 6.5 s (27 m), unchanged after notch filtering (2.7–6.6 s), single realization from in-house data. Control: no predicted-preview combination passed the gate, but **even perfect 8 s preview reduced RMS by only 0.4–0.9%**, and the loose stroke tier still hit saturation — control model lacked discriminative power, invalidating the control conclusion; Round 2 dispatched
10-14	Deep Research (--gemini)	Confirmed that real inflow preview for buildings/bridges is essentially a blank slate; identified genuine preview-control paper Marzbanrad 2004; compiled operating-logic table for Citicorp, Kyobashi Seiwa, Yokohama Landmark, Canton Tower, and PAFC (original texts unverified); found no forecast-switched operating procedures. Excluded: deriving 1–2 s from in-house 30 m (violates task brief), vehicle/passenger-boarding-bridge cases, and unreferenced "manual pre-warming several hours before a typhoon"
10-14	Lead original-text verification of Abdelaziz 2021	Verified (PDF p. 1 abstract): predictor ANN looks up flap angles from current wind conditions, steady 2D FSI, "preliminary results" show amplitude reduction up to 94%; this is "wind-condition scheduling", not inflow preview

Parameter Source Table

Parameter	Value	Source or "Assumed"	Places Checked
PAFC ATMD startup/lockout threshold	Not found in public literature	Grok snippet 0.01 m/s^2 / 2.2 m not adopted	Grok, DR, Crossref/OpenAlex abstracts (not accessible)
Citicorp TMD wake-up threshold	0.003 g for two cycles	Secondary citation via DR (vibrationdata/buffalo webpage), original text unverified	No hits in pages
Kyobashi Seiwa AMD stroke/force	±25 cm / 1 tf	Secondary citation via DR, unverified	—

Parameter	Value	Source or "Assumed"	Places Checked
Minimal experiment TPU scale / α	1/400, α=1/4, 1/6	TPU official case page (verified 10-13)	

Open Items and Next Steps

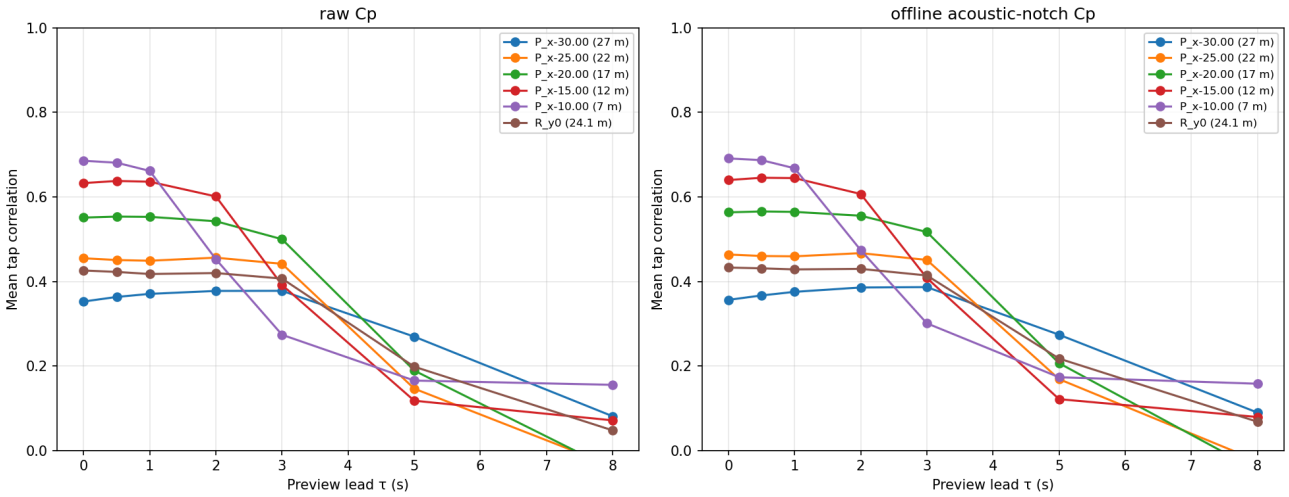
- Correlation–τ–distance table has been copied to the Digital Twin Wind Tunnel Root Lead, who archived it with no objections to the interpretation. After A2 was shelved, they will no longer specially export empty-domain building-location inflow dynamic pressure (job 1542028); if needed upon restart, request it directly from the Digital Twin Wind Tunnel Lead, who can provide it anytime.
- Hierarchy constraint (Root Lead 10-14): this is a Tier-2 node and will not open Tier-3 sub-projects; use tasks for subdivision; task briefs must not allow tool bots to delegate further.
- Grok report returns → read A3 Grok report → Deep Research brief (--gemini) → 3-reviewer review → define minimal GPU experiment candidates.
- In-house data reference: Silsoe LBM inflow → windward Cp correlation decays monotonically with upstream distance (0.67 at 7 m → 0.35 at 27 m), (in-house data, not literature).

Literature Basis

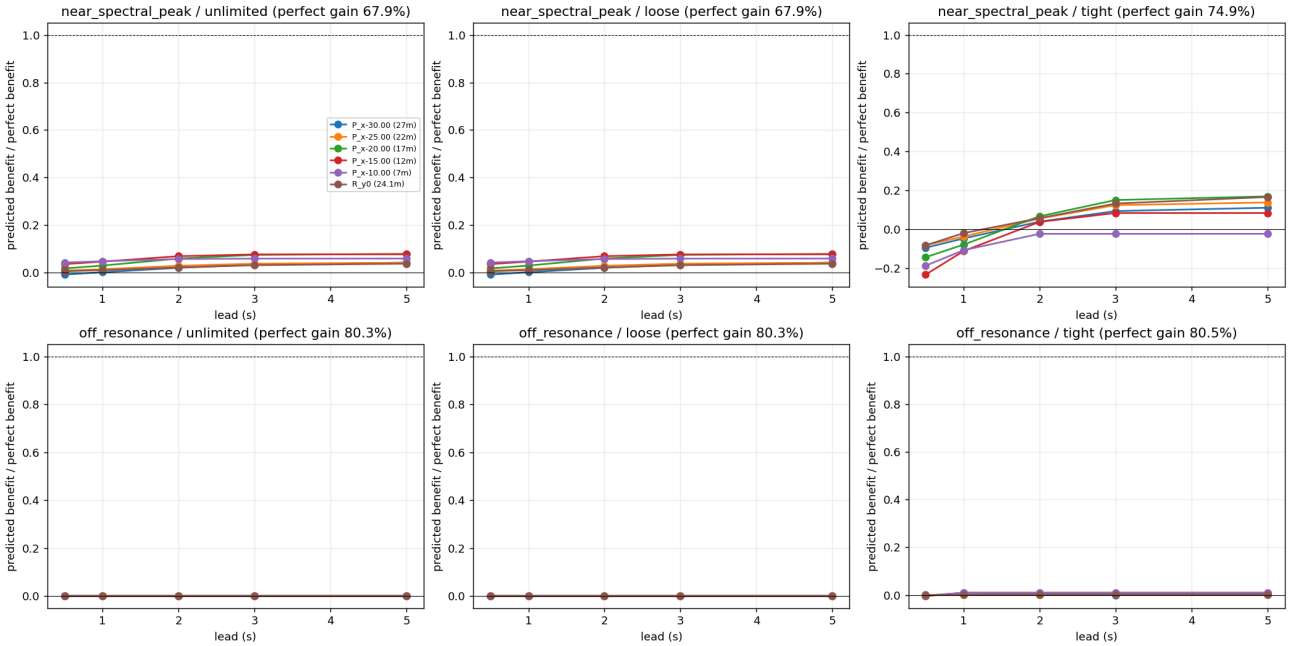
- Knowledge base (10-14 search): no cards on forecast/preview-driven structural control. Local full texts (pages) contain readable leads without cards yet: Abdelaziz et al. 2021 JWEIA 212:104601 smart facade control (Zotero U8L4S2VK, PDF p. 1); Korkmaz 2011 active structural control review (T8ADPIGM, PDF p. 18 = journal p. 2130, citing Abdel-Rohman 1983 aerodynamic appendages and Nissen 2004 bridge active aerodynamic stabilization); Kareem 1978 (2F6RHVB9, PDF p. 55 = journal p. 32). Cards to be created and verified.

Liang Hu Notes

Key figures



preview



control